

# MATHEMATICS

## EQUATIONALLY COMPLETE EXTENSIONS OF FINITE ALGEBRAS

BY

DANA SCOTT

(Communicated by Prof. A. HEYTING at the meeting of September 24, 1955)

It is the purpose of this note to prove the following:

*The set of identities of a finite algebra has only finitely many equationally complete extensions.*<sup>1)</sup>

In other words, the class of algebras generated by a finite algebra includes only finitely many equationally complete classes of algebras. In the course of demonstration of this theorem we shall develop a characterization of those algebras, finite or not, which are equationally complete. For a finite algebra the condition yields an effective test of whether the algebra is equationally complete; and, if it is not, then the method still enables us to compute the number of possible equationally complete extensions. The proof uses in an essential way the notion of a polynomial algebra, whose definition we now recall. For simplicity we consider only algebras  $\mathfrak{A} = \langle A, +_{\mathfrak{A}} \rangle$  with a single binary operation, but all the results of this paper hold for algebras with any finite number of finitary operations.

**Definition 1.** Consider all terms, or polynomials, built up from the variables ' $x_0$ ', ' $x_1$ ', ..., ' $x_{n-1}$ ', parentheses, and the binary operation symbol '+'. For such a term  $\alpha$ , let  $E(\alpha)$  be the class of all terms  $\beta$  such that the equation ' $\alpha = \beta$ ' holds in  $\mathfrak{A}$ . The classes  $E(\alpha)$  form an algebra  $\mathfrak{P}_n[\mathfrak{A}]$  whose binary operation is defined by the formula

$$E(\alpha) +_{\mathfrak{P}_n[\mathfrak{A}]} E(\beta) = E(\ulcorner \alpha + \beta \urcorner).$$

$\mathfrak{P}_n[\mathfrak{A}]$  is called the *polynomial algebra with  $n$  generators over  $\mathfrak{A}$* .<sup>2)</sup>

We state as lemmas several general properties of polynomial algebras which are either known or very easily proved.

**Lemma 1.** *The identities of  $\mathfrak{P}_n[\mathfrak{A}]$  include those of  $\mathfrak{A}$ .*

<sup>1)</sup> For notation and terminology see KALICKI-SCOTT [3]. It is interesting to compare the arguments here with those in TARSKI [5], where a condition is given for a system of sentential calculus to have one complete extension. The author has obtained the analogous of the theorems and methods of this note for systems of sentential calculi and thus can offer a generalization of TARSKI's result.

<sup>2)</sup> This definition is essentially that of BIRKHOFF [1], pp. viii-ix.

Lemma 2. *The identities in  $n$  variables of  $\mathfrak{P}_n[\mathfrak{A}]$  are the same as those of  $\mathfrak{A}$ .*

Lemma 3. *The identities in  $n$  variables of an algebra  $\mathfrak{B}$  are the same as those of  $\mathfrak{A}$  if and only if  $\mathfrak{P}_n[\mathfrak{B}]$  is isomorphic to  $\mathfrak{P}_n[\mathfrak{A}]$ .*

Lemma 4.  *$\mathfrak{P}_n[\mathfrak{P}_n[\mathfrak{A}]]$  is isomorphic to  $\mathfrak{P}_n[\mathfrak{A}]$ .*

Lemma 5. *The identities in  $n$  variables of  $\mathfrak{B}$  include those of  $\mathfrak{A}$  if and only if  $\mathfrak{P}_n[\mathfrak{B}]$  is a homomorphic image of  $\mathfrak{P}_n[\mathfrak{A}]$ .*

Lemma 6. *If  $\mathfrak{A}$  is finite, then  $\mathfrak{P}_n[\mathfrak{A}]$  is finite.*

Lemma 7. *If  $n > 1$  and  $\mathfrak{A}$  has at least two elements, then  $\mathfrak{P}_n[\mathfrak{A}]$  has at least two elements.*

Lemma 8. *If  $n > 1$  and  $\mathfrak{A}$  is equationally complete, then  $\mathfrak{P}_n[\mathfrak{A}]$  is equationally complete, and its identities are the same as those of  $\mathfrak{A}$ .*

We are now in a position to prove our main theorem in a somewhat more general form than that of the result stated above, which, however, follows at once by virtue of Lemma 6.

Theorem 1. *If  $n > 1$  and  $\mathfrak{P}_n[\mathfrak{A}]$  is finite, then there are only finitely many equationally complete classes of algebras included in the class generated by  $\mathfrak{A}$ .*

Proof: Let  $\mathbf{L}$  be an equationally complete class included in the class  $\mathbf{K}$  generated by  $\mathfrak{A}$ . Let  $\mathfrak{B}$  be any member of  $\mathbf{L}$  with at least two elements.  $\mathfrak{B}$  is in the class generated by  $\mathfrak{A}$ , and hence the identities of  $\mathfrak{B}$  include those of  $\mathfrak{A}$ . By Lemma 5,  $\mathfrak{P}_n[\mathfrak{B}]$  is a homomorphic image of  $\mathfrak{P}_n[\mathfrak{A}]$ . Since  $\mathfrak{B}$  is equationally complete,  $\mathfrak{B}$  and  $\mathfrak{P}_n[\mathfrak{B}]$  have the same identities in view of Lemma 8; whence  $\mathfrak{P}_n[\mathfrak{B}]$  is in  $\mathbf{L}$  and generates  $\mathbf{L}$ .<sup>3)</sup> If  $\mathfrak{C}$  is in  $\mathbf{L}$  and has at least two elements, then  $\mathfrak{B}$  and  $\mathfrak{C}$  have the same identities, and  $\mathfrak{P}_n[\mathfrak{B}]$  is isomorphic to  $\mathfrak{P}_n[\mathfrak{C}]$  by Lemma 3. Thus we have associated with  $\mathbf{L}$  a well determined isomorphism type of homomorphic images of  $\mathfrak{P}_n[\mathfrak{A}]$  which in turn determines  $\mathbf{L}$ . Clearly then to distinct equationally complete classes included in  $\mathbf{K}$  there correspond distinct types of homomorphic images of  $\mathfrak{P}_n[\mathfrak{A}]$ .  $\mathfrak{P}_n[\mathfrak{A}]$  is finite and can have only finitely many distinct types of homomorphic images; hence, Theorem 1 is proved.

In proving this theorem, we have actually obtained far more information than was strictly needed. This additional material is stated and proved in Theorem 2; the formulation is facilitated by the following definition.

Definition 2.  $\mathfrak{A}$  is  $\mathfrak{P}_n$ -simple if and only if  $\mathfrak{A}$  has at least two elements, and, for every homomorphic image  $\mathfrak{B}$  of  $\mathfrak{P}_n[\mathfrak{A}]$ , either  $\mathfrak{B}$  is a one element algebra or  $\mathfrak{P}_n[\mathfrak{B}]$  is isomorphic to  $\mathfrak{P}_n[\mathfrak{A}]$ .

<sup>3)</sup> See KALICKI-SCOTT [3], Theorems 1.6 and 1.7.

**Theorem 2.** *If  $n > 1$ , then  $\mathfrak{A}$  is equationally complete if and only if  $\mathfrak{A}$  is  $\mathfrak{P}_n$ -simple and is in the class generated by  $\mathfrak{P}_n[\mathfrak{A}]$ .<sup>4)</sup>*

**Proof:** [Necessity] Assume that  $\mathfrak{A}$  is equationally complete. By definition  $\mathfrak{A}$  has at least two elements. By Lemma 8,  $\mathfrak{A}$  and  $\mathfrak{P}_n[\mathfrak{A}]$  have the same identities, and hence  $\mathfrak{A}$  is in the class generated by  $\mathfrak{P}_n[\mathfrak{A}]$ . Let  $\mathfrak{B}$  be a homomorphic image of  $\mathfrak{P}_n[\mathfrak{A}]$  with at least two elements. Now the identities of  $\mathfrak{B}$  include those of  $\mathfrak{P}_n[\mathfrak{A}]$  which are the same as those of  $\mathfrak{A}$ . Since  $\mathfrak{A}$  is equationally complete,  $\mathfrak{A}$  and  $\mathfrak{B}$  must have the same identities. By Lemma 3,  $\mathfrak{P}_n[\mathfrak{B}]$  and  $\mathfrak{P}_n[\mathfrak{A}]$  are isomorphic. Hence,  $\mathfrak{A}$  is  $\mathfrak{P}_n$ -simple, and this completes the proof in one direction.

[Sufficiency] Assume that  $\mathfrak{A}$  is  $\mathfrak{P}_n$ -simple and is in the class generated by  $\mathfrak{P}_n[\mathfrak{A}]$ . Thus, the identities of  $\mathfrak{A}$  include those of  $\mathfrak{P}_n[\mathfrak{A}]$ , but by Lemma 1 the inclusion holds in the other direction; hence,  $\mathfrak{A}$  and  $\mathfrak{P}_n[\mathfrak{A}]$  have the same identities. Let  $\mathfrak{B}$  be an equationally complete algebra whose identities include those of  $\mathfrak{A}$ .<sup>5)</sup> By Lemma 5,  $\mathfrak{P}_n[\mathfrak{B}]$  is a homomorphic image of  $\mathfrak{P}_n[\mathfrak{A}]$ ; however,  $\mathfrak{B}$  has at least two elements and so does  $\mathfrak{P}_n[\mathfrak{B}]$  in view of Lemma 7. Thus, since  $\mathfrak{A}$  is  $\mathfrak{P}_n$ -simple,  $\mathfrak{P}_n[\mathfrak{P}_n[\mathfrak{B}]]$  is isomorphic to  $\mathfrak{P}_n[\mathfrak{A}]$ . By Lemma 4,  $\mathfrak{P}_n[\mathfrak{B}]$  is isomorphic to  $\mathfrak{P}_n[\mathfrak{P}_n[\mathfrak{B}]]$ , and by Lemma 8,  $\mathfrak{B}$  and  $\mathfrak{P}_n[\mathfrak{B}]$  have the same identities. Hence all these algebras have the same identities, and  $\mathfrak{A}$  is therefore equationally complete. Theorem 2 is thus proved.

Theorem 2 yields an effective procedure for determining whether a finite algebra is equationally complete. Thus, if  $\mathfrak{A}$  is finite, then  $\mathfrak{P}_2[\mathfrak{A}]$  is finite and can indeed be constructed in a finite number of steps. To check whether  $\mathfrak{A}$  is  $\mathfrak{P}_2$ -simple, one has to examine the finite number of distinct types of homomorphic images  $\mathfrak{B}$  of  $\mathfrak{P}_2[\mathfrak{A}]$ , and, in case  $\mathfrak{B}$  has at least two elements, to see that the two finite algebras  $\mathfrak{P}_2[\mathfrak{B}]$  and  $\mathfrak{P}_2[\mathfrak{A}]$  are isomorphic. To check whether  $\mathfrak{A}$  is in the class generated by  $\mathfrak{P}_2[\mathfrak{A}]$  is to check whether  $\mathfrak{A}$  and  $\mathfrak{P}_2[\mathfrak{A}]$  have the same identities; and, if  $m$  is the maximum of the number of elements of  $\mathfrak{A}$  and of  $\mathfrak{P}_2[\mathfrak{A}]$ , then this is the same as checking whether the two finite algebras  $\mathfrak{P}_m[\mathfrak{A}]$  and  $\mathfrak{P}_m[\mathfrak{P}_2[\mathfrak{A}]]$  are isomorphic<sup>6)</sup>. Thus, all the checks can be made in a finite number of steps.

If  $\mathfrak{A}$  is finite but not equationally complete, then we can still determine the exact number of equationally complete classes included in the class generated by  $\mathfrak{A}$ . Since these classes are in a one-one correspondence with the distinct types of homomorphic images  $\mathfrak{B}$  of  $\mathfrak{P}_2[\mathfrak{A}]$ , where  $\mathfrak{B}$  is  $\mathfrak{P}_2$ -simple and is isomorphic to  $\mathfrak{P}_2[\mathfrak{B}]$ , we can find their exact number by

---

<sup>4)</sup> Since polynomial algebras can be defined in a purely algebraic way as is done in McKINSEY-TARSKI [4] pp. 184–188, this theorem then yields a purely algebraic characterization of equational completeness.

<sup>5)</sup> See KALICKI-SCOTT [3], Theorem 1.4.

<sup>6)</sup> For more detailed information about checking whether two finite algebras have the same identities, see KALICKI [3].

examining the finite number of types of homomorphic images of the finite algebra  $\mathfrak{P}_2[\mathfrak{A}]$ .<sup>7)</sup> Obviously there is no claim as to the practicality of these methods.

We can summarize the results of this note in the following way: the study of the degree of equational completeness of an algebra  $\mathfrak{A}$  reduces to the study of that of  $\mathfrak{P}_2[\mathfrak{A}]$ , since the equationally complete extensions of the identities of  $\mathfrak{A}$  are the same as those for  $\mathfrak{P}_2[\mathfrak{A}]$ . Further, the complete extensions for  $\mathfrak{P}_2[\mathfrak{A}]$  are all represented by certain special homomorphic images of that algebra. Thus results the possibility of an exhaustive description of equational completeness in the finite case.

*University of California, Berkeley*

#### BIBLIOGRAPHY

1. BIRKHOFF, G., *Lattice Theory*, American Mathematical Society Colloquium Publications, 25, xiii + 283 pp. (revised, New York, 1948).
2. KALICKI, J., *On comparison of finite algebras*, Proc. Amer. Math. Soc., 3, 36-40 (1952).
3. ——— and D. SCOTT, *Equational completeness of abstract algebras*, Indag. Math., 17, 650-659 (1955).
4. MCKINSEY, J. C. C. and A. TARSKI, *The algebra of topology*, Annals of Math., 45, 141-191 (1944).
5. TARSKI, A., *Über die Erweiterungen der unvollständigen Systeme des Aussagenkalküls*, Ergebnisse eines Mathematischen Kolloquiums, 7, 51-57 (1935).

---

<sup>7)</sup> Thus, if  $n$  is the number of elements of  $\mathfrak{A}$  and  $m = n^2$ , then  $2^{m^m}$  is a very generous upper bound for the number of equationally complete extensions of the identities of  $\mathfrak{A}$ . Notice that in these remarks  $\mathfrak{P}_n[\mathfrak{A}]$  has been used only for  $n = 2$ ; also notice that Theorems 1 and 2 use the condition  $n > 1$ . The conditions needed for using  $n = 1$  could have been formulated, but it did not seem worthwhile.